Field accuracy is not heat accuracy: wall-flux-constrained surrogates for steady two-dimensional non-Boussinesq convection

Abstract

Paired Boussinesq and compressible simulations are expensive, and public data sets fix the relative contrast $\varepsilon = \Delta T/T_0$, while surrogates are ranked by L_2 field error. For steady two-dimensional buoyancy-driven enclosure convection, this metric can rank a model above doing nothing on wall heat transfer. On 200 held-out classical-cavity samples, a prior with 0.09% median field error misses the both-wall integral Nusselt number by 2.0% on average (1.4–2.9% across seeds); in 43% of samples on average (23–82% across seeds) field error is below 1% but Nu error above 2%. A 22-sample random-wall replication at $\varepsilon \approx 1.2$ gives 0.47% median field error and 4.04% median Nu error, with 66% on average (50–77% across seeds) there. Under the training stencil, a wall-flux penalty reduces these Nu errors to 0.01% and 0.58%, relative to the 64^2 compressible target; four/eight-point stencils give 0.3–0.9%, overlapping returning X (0.4–0.9%). Six-point local errors are 0.08% and 1.14%, and median field error moves to 0.10% and 0.49%. At $\varepsilon = 1.2$ the hot-wall integral Nu changes by 0.67% while the L_1 local-profile change is 4.25% (Table 2's 5.6% uses a different, mean-absolute denominator). The correction factorises as $D = \varepsilon P(X) + O(\varepsilon^2)$ across 627 paired solutions with $\varepsilon = 0.02$–1.9; the velocity channels follow $R = A\varepsilon^m$ over the full window with $m = 1.023 \pm 0.018$ and 0.995 ± 0.019. A quadratic refinement gives $\kappa = 0.70$ and 0.83 (point estimates); A is a functional of the flow. A public single-contrast data set then works at $\varepsilon = 0.2$–1.9 within its boundary family: a plain model trained at $\varepsilon_0 = 1.2$ and rescaled by $\varepsilon/\varepsilon_0$ agrees with the factorized target to 0.2% relative, so applying ε outside the learned correction, not the residual parameterisation, carries the transfer. At $\varepsilon = 1.9$ the integral Nu change is -5.1% at $Ra = 10^4$ and -1.5 to -1.8% at $Ra = 10^5$–10^6. Inference takes 6.3 ms; replacing the compressible member saves 2.9× per pair and 13× for a 20-flow × 5-contrast scan.

Keywords: natural convection, Boussinesq approximation, low-Mach compressible flow, physics-informed surrogate, data reuse, wall heat flux

Nomenclature

Roman		Greek	
A	prefactor of $R = A\varepsilon^m$	ΔT	wall-to-wall temperature contrast
D	correction $Y - X$	ε	relative contrast, $\Delta T/T_0$
Fr	Froude number, $Fr^2 = \Delta T/T_0$	θ	normalised temperature
m	scaling exponent	κ	curvature in $R = A\varepsilon(1 + \kappa\varepsilon)$
Ma	Mach number	ρ	density
Nu	Nusselt number	χ	base-flow advection strength
p	pressure		
E, e	total and internal energy	$\boldsymbol{P}_{\mathrm{work}}$	pressure-work tensor
P, Q	first- and second-order correction fields		
Pr	Prandtl number		
R	relative correction, $\|Y - X\|/\|Y\|$		
Ra	Rayleigh number		
Re	Reynolds number, $(Ra/Pr)^{1/2}$		
T	temperature, T_0 its reference value		
$\boldsymbol{u}$	velocity, (u_x, u_y)		
$\hat{u}$	peak Boussinesq speed		
X, Y	Boussinesq and compressible solutions	$\mathcal{L}$	linearised low-Mach operator at $O(\varepsilon)$

1. Introduction

Steady natural convection is a workhorse of thermal engineering and it is expensive. A converged pair of two-dimensional Boussinesq and compressible solves on a 64^2 mesh costs 34 s of wall clock on an eight-way desktop (Sec. 4.4); three-dimensional problems cost hours to days. Here the scaling law and the speed-up are measured in two dimensions; the three-dimensional pairs are used only to check the exponent. Surrogate modelling removes this cost by learning the map from a cheap model to an expensive one [1, 2]. For convection the two models are not arbitrary fidelities: they are the Oberbeck–Boussinesq equations and the low-Mach compressible equations. Boussinesq is exact only as the relative temperature contrast vanishes; the difference is $O(\beta\Delta T)$ and has been characterised asymptotically for decades [3, 4].

The numerical literature offers several ways to reduce the cost. Reduced-order models replace the discretised equations by a small number of modes [5]; graph convolutional networks exploit the mesh [6, 7]; operator learning removes the mesh-resolution constraint [8, 9, 10], with applications to sparse-field reconstruction [11] and heat-pipe startup [12]. Physics-informed neural networks take the opposite route and solve one instance by minimising a residual [13, 14, 15, 16]; reviews of the numerical analysis and the application landscape are given by De Ryck and Mishra [17] and Luo et al. [18]. These methods reduce the cost of a single query, but they still need paired data at the operating conditions of interest.

Non-Oberbeck–Boussinesq effects have been measured under strong temperature contrast [19], in cryogenic helium [20], and in water-filled cavities [21]; at large contrast they can grow cold-corner vortices and reverse the

large-scale circulation [22]. The heat-transfer and large-scale dynamics of Rayleigh–Bénard convection are reviewed by Ahlers et al. [23]. Low-Mach formulations for enclosed thermal convection have been developed and compared with the Boussinesq limit [24, 25], and the rigorous derivation of the Oberbeck–Boussinesq system has been revisited [26]. These studies fix the contrast and compare the two solutions. They do not extract its dependence as a reusable scaling.

The practical difficulty appears when a surrogate is trained on a paired data set at a single contrast. Menglik et al. [27] released 4000 paired Boussinesq→compressible steady solutions in two dimensions, with three held-out splits, at one value of $\varepsilon = \Delta T/T_0 = 1.2$. A network trained on that release learns the correction at $\varepsilon = 1.2$. At another contrast the Boussinesq input is unchanged, because the Boussinesq equations do not contain ε, while the correction that must be added scales with ε. The network has no signal with which to rescale what it memorised. The usual remedy is a second paired data set, which pays again the cost the surrogate was meant to remove.

This paper removes that barrier. The correction between the two members factorises. An ε-expansion of the compressible member gives the Boussinesq system at $O(1)$ and a linear, ε-independent operator at $O(\varepsilon)$. Hence

$$D = Y - X = \varepsilon P(X) + \varepsilon^2 Q(X) + \cdots , \tag{1}$$

so a surrogate only has to learn the spatial shape P, and ε can be applied outside the network. The same expansion identifies the parameter that controls the gap: $Fr^2 = \varepsilon$ identically in the non-dimensionalisation used by the released data and by our solver, while Ma enters the retained equations only at $O(Ma^2)$. The break-symmetry axis is therefore one-dimensional.

We test the consequence on 627 paired solutions in the unit square, on a second geometry, on a second Prandtl number, on 33 three-dimensional pairs, and on two public data sets generated independently of our solver. The law holds for the velocity and pressure channels; the temperature channel has a flow-dependent amplitude that collapses on the peak Boussinesq speed. The scaling law makes a single-contrast public data set reusable within its boundary-condition family, and the limit is equally clear: the law transfers the ε dependence, not the flow-dependent prefactor.

The central result, however, is not the law by itself. The correction is largely a redistribution of wall flux: at $\varepsilon = 1.2$ the integral Nusselt number of the classical cavity changes by 0.67% on the hot wall while the local wall flux changes by 4.25% in L_1. A surrogate can therefore reproduce the temperature field to 0.47% in relative L_2 and still miss the integral Nu by 4%. On the 22 held-out random-wall samples at $\varepsilon = 1.1$–1.3, 66% of the field-only predictions on average across five seeds have a field error below 1% and a Nu error above 2%; on the 200 cavity samples, 43% on average

across three seeds do. Relative L_2 is not a sufficient selection metric for a heat-transfer surrogate. A wall-flux penalty in the training loss reduces the heat-transfer bias at a small cost in field error.

The contributions are:

1. a per-sample demonstration that relative L_2 field accuracy does not imply Nusselt-number accuracy, with a mechanism: the correction redistributes the wall-normal temperature gradient while the integral almost cancels;
2. a wall-flux training penalty that reduces the both-wall Nu error from 4.0% to 0.58% on the random-wall replication (0.01% on the classical cavity), and the local wall-flux error to 1.2% or below;
3. a quantitative law for the Boussinesq–compressible gap, $R = A\varepsilon^m$ with m indistinguishable from one over the full window and a positive quadratic curvature, calibrated over two decades of ε and tested across geometry and fluid, with a 33-pair three-dimensional consistency check, together with a within-family data-reuse rule. The factorized target is one implementation; applying the known ε scaling is what makes the data reusable, so the factorisation, not the network, is the reusable object.

The main sweep is two-dimensional and steady. The three-dimensional check uses 33 pairs at three Rayleigh numbers and one resolution; it tests the exponent, not a three-dimensional design workflow. The scope is stated in Sec. 5.

2. Physical scaling

2.1. Two members and one contrast parameter

Each sample is a paired steady solution of one configuration. The high-fidelity member Y is the low-Mach compressible formulation in $(\boldsymbol{u}, T, \rho, p)$:

$$\nabla \cdot (\rho \boldsymbol{u}) = 0, \tag{2}$$

$$\nabla \cdot (\rho \boldsymbol{u} \otimes \boldsymbol{u}) = -\nabla p + \frac{1}{Re}\nabla \cdot \boldsymbol{\sigma} - \frac{1}{Fr^2}\rho \boldsymbol{g}, \tag{3}$$

$$\nabla \cdot (\rho E \boldsymbol{u}) = \frac{\gamma}{Re\,Pr}\nabla^2 e + \nabla \cdot (\boldsymbol{P}_{\text{work}}\boldsymbol{u}) + O(Ma^2), \tag{4}$$

$$\rho e = \gamma Ma^2 p + 1, \qquad Re = (Ra/Pr)^{1/2}. \tag{5}$$

The low-fidelity member X is the Boussinesq system in $(\boldsymbol{u}, T, p)$ on the same mesh:

$$\nabla \cdot \boldsymbol{u} = 0, \tag{6}$$

$$\boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\nabla p + \left(\frac{Pr}{Ra}\right)^{1/2} \nabla^2 \boldsymbol{u} + \theta \boldsymbol{g}, \tag{7}$$

$$\boldsymbol{u} \cdot \nabla \theta = (Ra\, Pr)^{-1/2} \nabla^2 \theta, \qquad \theta = \frac{T - T_0}{\Delta T}. \tag{8}$$

Both members use Taylor–Hood-type mixed elements, damped Newton iterations and a MUMPS factorisation. They start from the same initial field and the same wall-temperature seed, so they follow the same branch. Left and right walls are Dirichlet; top and bottom walls are Dirichlet when their Fourier dial is nonzero and adiabatic otherwise; velocity is no-slip on all walls. The local Nusselt profile is $-\partial\theta/\partial n$ from a six-point one-sided stencil, and the integral Nu is its wall average. The classical differentially heated cavity is the one-mode member of the same family.

The generator draws Ra and ε independently. At fixed Ra, changing ε changes $Fr = \sqrt{\varepsilon}$, the buoyancy scale, while Re, Pr and γ stay fixed. The Boussinesq member sees only Ra and the thermal boundary data. This is a statement about the dimensionless parameterisation; a fixed apparatus follows the one-dimensional slice $Ra \propto \varepsilon$, while the (Ra, ε) rectangle contains that slice.

Two simplifications follow from the equations. First, $Fr^2 = \beta\Delta T = \Delta T/T_0$ identically in this non-dimensionalisation, so Fr is not a second free axis. Second, Ma appears only at $O(Ma^2)$ in the retained low-Mach system. A 24-fold change in Ma at fixed Ra, T_0 and ΔT leaves the measured correction unchanged to a relative 10^{-5}, which is the precision of the solve. The free parameter is ε.

The Boussinesq solution is ε-blind to solver precision. Neither T_0 nor ΔT appears alone in Eqs. (6)–(8); with the thermal data prescribed in θ and Ra fixed, X cannot depend on ε. Across 20 configurations $\times$ 5 contrasts, the maximum relative deviation of X is 9.77×10^{-8}, the solver's convergence floor. The amplitude of the correction is thus set by a variable that is absent from the input. The spatial shape remains a function of X and is learnable.

2.2. Why the correction factorises

Write $T = T_0(1 + \varepsilon\theta)$ with $\theta = O(1)$ and use $\rho = 1/T$. The buoyancy source becomes

$$\frac{\rho \boldsymbol{g}}{Fr^2} = \frac{\rho \boldsymbol{g}}{\varepsilon} = \frac{\boldsymbol{g}}{\varepsilon} - \theta \boldsymbol{g} + \varepsilon\theta^2 \boldsymbol{g} + \cdots . \tag{9}$$

The constant $\boldsymbol{g}/\varepsilon$ is balanced by a hydrostatic pressure p_{-1}/ε with $\nabla p_{-1} = -\boldsymbol{g}$, which drops out at the next order. The $O(1)$ balances are

$$\nabla \cdot \boldsymbol{u}_0 = 0, \tag{10}$$

$$\boldsymbol{u}_0 \cdot \nabla \boldsymbol{u}_0 = -\nabla \tilde{p}_0 + Re^{-1} \nabla^2 \boldsymbol{u}_0 + \theta_0 \boldsymbol{g}, \tag{11}$$

$$\boldsymbol{u}_0 \cdot \nabla \theta_0 = (Re\, Pr)^{-1} \nabla^2 \theta_0, \tag{12}$$

which is term-by-term the Boussinesq system solved as X, with $Re^{-1} = (Pr/Ra)^{1/2}$. The correction at $O(\varepsilon)$ is driven by the density modulation:

$$\nabla \cdot \boldsymbol{u}_1 = \nabla \cdot (\theta_0 \boldsymbol{u}_0), \tag{13}$$

$$\nabla \cdot (\boldsymbol{u}_1 \otimes \boldsymbol{u}_0 + \boldsymbol{u}_0 \otimes \boldsymbol{u}_1 - \theta_0 \boldsymbol{u}_0 \otimes \boldsymbol{u}_0) = -\nabla \tilde{p}_1 + Re^{-1} \nabla^2 \boldsymbol{u}_1 + \theta_1 \boldsymbol{g} - \theta_0^2 \boldsymbol{g}. \tag{14}$$

The mass equation is inhomogeneous, so the operator is not the Boussinesq operator linearised about X; it is the linearisation of the retained low-Mach system about X, with the mass constraint carried inside. Its only singular mode is the constant pressure, and the compatibility condition holds because $\boldsymbol{u}_0 \cdot \boldsymbol{n} = 0$ on the walls. Since the operator is built from X and contains no explicit ε,

$$D = -\varepsilon \mathcal{L}^{-1} S_1[X] - \varepsilon^2 \mathcal{L}^{-1} S_2[X] + \cdots = \varepsilon P(X) + \varepsilon^2 Q(X) + \cdots . \tag{15}$$

Five predictions follow: $R \to 0$ as $\varepsilon \to 0$; $R \propto \varepsilon$ with a vanishing intercept; the shape of D does not rotate with ε; $\nabla \cdot \boldsymbol{u}_1 = \nabla \cdot (\theta_0 \boldsymbol{u}_0)$; and the ε^1 term carries most of the correction energy. All five hold numerically (Supplementary Material). The ε^1 term carries 86% of the correction energy, $\varepsilon^1 + \varepsilon^2$ carries 99.0%, and three terms carry 99.9% over the sampled range. The divergence identity has cosine 0.994 for $\varepsilon < 0.6$ and 0.901 at $\varepsilon = 1.9$; the drop is the footprint of the ε^2 term.

The expansion is valid for $\varepsilon < 2$, because the cold wall $T_c = T_0 - \Delta T/2$ must remain positive. At $\varepsilon = 1.9$, $T_c/T_0 = 0.05$, so the density expansion is close to its limit. This is the quantitative boundary of the law, not a fitting choice.

2.3. The measured law

The calibration set contains 600 random-boundary samples spanning 134 wall-mode families, with $Ra \sim \log U[10^2, 10^6]$ and $\varepsilon \sim U[0.2, 1.9]$, plus 27 low-contrast checks at $\varepsilon = 0.02, 0.05, 0.10$ and $Ra = 10^3, 10^4, 10^5$, with three independent seeds per condition. The $600 + 27 = 627$ points enter the fit; the low-contrast points anchor the $\varepsilon \to 0$ end. The relative correction is $R = \|Y - X\|/\|Y\|$.

Table 1 and Fig. 1 give the result. For u_x and u_y the free power law returns $m = 1.023 \pm 0.018$ and 0.995 ± 0.019. The pressure channel gives

$m = 1.009 \pm 0.015$ in the fixed gauge described below; it is a numerical observation, not a gauge-invariant exponent. The intercepts are consistent with zero. A quadratic fit $R = a\varepsilon + c\varepsilon^2 + b$ gives $c > 0$ in all four channels with point estimates $\kappa_{\text{fit}} = c/a = 0.70/0.83/0.42/0.73$ for $u_x/u_y/\theta/p$ (no confidence intervals are available). A compact description is therefore $R = A\varepsilon(1 + O(\varepsilon))$ rather than an exact power. Fitting only $\varepsilon \geq 0.8$ returns $m = 1.23$–1.32, the window average of the same curvature.

Table 1: Power-law fit $R = A\varepsilon^m$ over the full two-decade window (627 samples). The intercept is fitted separately on $\varepsilon \leq 0.5$. The pressure row uses the fixed gauge of Sec. 2.5; its exponent is not gauge-invariant. θ is not included: its free power-law fit is not identifiable, and its amplitude is discussed separately.

channel	A	m	R^2	intercept ($\varepsilon \leq 0.5$)
u_x	0.147	1.023 ± 0.018	0.84	$+0.0003 \pm 0.0047$
u_y	0.193	0.995 ± 0.019	0.82	$+0.0030 \pm 0.0062$
p	0.148	1.009 ± 0.015	0.87	$+0.0016 \pm 0.0030$

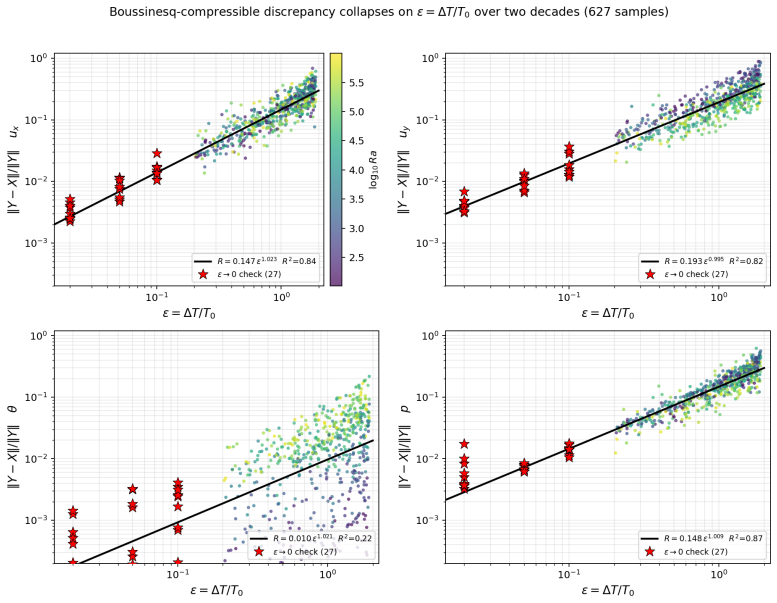


Figure 1: The Boussinesq–compressible discrepancy collapses on $\varepsilon = \Delta T/T_0$ for u_x, u_y and p over two decades. The θ panel scatters because its amplitude depends on the flow; it collapses on the peak Boussinesq speed (Supplementary Material). Points: 600 random-boundary samples. Stars: 27 low-contrast checks. Lines: power-law fits of Table 1 for u_x, u_y and p; the θ line is a free fit ($R^2 = 0.22$) and is not part of the common-exponent claim.

The temperature channel needs a flow-dependent amplitude. Its free power law has $R^2 = 0.22$ and a slope that moves from 1.02 over the full

window to 1.67 on the 104-configuration subset; the fitted amplitude is not predictive. This is not a failure of the expansion. Writing $R_\theta/\varepsilon = c\hat{u}^a$, where $\hat{u}$ is the peak Boussinesq speed available from X, collapses six data sets from two independent generators: $a = 1.86$–1.98 and $c = 0.54$–0.69 for random-wall families, $a = 2.36$ and $c = 2.06$–2.10 for the classical cavity. The classical cavity has a single strong circulation with $\hat{u}$ 41% above the random-wall median and 56% above ours. The same reading explains the residual scatter in u_x, u_y and p: the prefactor is a functional of the flow, and the flow is in the input X.

2.4. What changes and what does not

The exponent is stable; the prefactor is not. In a 2:1 cavity with 104 samples, the ε-only fit gives $m = 1.09/1.16/0.93/1.08$ for $u_x/u_y/\theta/p$, within 0.3–1.8 combined standard errors of the unit-square values. The prefactor moves from 0.147/0.193/0.010/0.148 to 0.175/0.175/0.008/0.222. Changing the fluid from $Pr = 0.71$ to $Pr = 4.38$ in 104 strictly paired configurations leaves the exponent within 1.5% and the paired median R within 5%; the prefactor moves by at most 3.3% except for pressure (6.1%). A 33-sample three-dimensional check at $Ra = 10^4$, 3×10^4 and 10^5 ($res = 16$) returns exponents for u_x, u_y and p of 1.09/1.05/1.14, 1.03/1.04/1.09 and 1.01/1.01/1.09, all within 0.15 of unity. The prefactor varies by factors of 2–13 across the three wall types.

The practical reading is simple. The law supplies the ε dependence. It does not supply the flow dependence. A surrogate must therefore take X as input rather than a table of A. This is also why transfer across boundary-condition families fails: the model has learned one family's prefactor, not a universal constant.

2.5. The pressure channel

The two members store pressure in different conventions. The compressible raw pressure carries a hydrostatic ramp 12 times larger than its dynamic part, and the two weak forms use opposite signs. Comparing raw pressures measures the conventions, not the flow. We subtract the mean, remove a least-squares linear fit in the gravity direction, and flip the sign of the Boussinesq member. Under this gauge the per-sample median of $|X_p|/|Y_p|$ is 0.960 and the cosine is +0.991, against 0.929 and +0.986 for the public training set. The pressure channel then follows the same law as the velocities, with $A = 0.148$ and $m = 1.009$. All pressure numbers use this fixed gauge; the exponent is not gauge-invariant.

3. Surrogate design

3.1. Residual learning and the factorised target

All surrogates use residual learning,

$$\hat{Y} = X + N(X), \tag{16}$$

with a U-Net of width 32 or an FNO with 12 modes, width 32 and four layers (1.19M parameters). Training uses Adam with cosine annealing, 200 epochs and batch 16. Inputs and targets are scaled per channel by training-set standard deviations.

The prior changes the target, not the backbone. Instead of learning D, the network learns $P = D/\varepsilon$ at the available contrast and the model returns

$$\hat{Y} = X + \varepsilon N(X). \tag{17}$$

We compare five variants:

- A: ε-blind, target D;
- B: ε supplied as a constant input channel, target D;
- P1: ε supplied as input, target D/ε, multiplied outside;
- P2: as P1 with a learned scalar curvature, $\varepsilon(1 + \kappa\varepsilon)$;
- P3: as P1 with no ε input, so the network sees only X.

P1 is the most accurate form when ε is known at training time. P3 is the form that can be trained on a single-contrast data set and applied at another contrast; it must infer the prefactor from the flow. The distinction is the difference between a model that interpolates in ε and one that reuses a data set.

3.2. Wall-flux loss

Field accuracy does not guarantee heat-transfer accuracy. For applications we add a relative L_1 penalty on the wall-normal gradient,

$$\mathcal{L} = \mathcal{L}_{\text{field}} + \lambda \frac{\sum_{\text{wall}} |\hat{q} - q|}{\sum_{\text{wall}} |q|}, \qquad q = -\frac{\partial \theta}{\partial n}, \tag{18}$$

with $\lambda = 1$ and a six-point one-sided stencil. The same models are evaluated with two-, four-, six- and eight-point stencils to separate the loss from the estimator.

3.3. Data, splits and metrics

The main data set is the 600-sample random-boundary scan, with the 27 low-contrast checks used for the law. The paired scan contains 20 configurations × 5 contrasts; within a configuration only ε changes, so X is identical and the error of any ε-blind model is bounded exactly. The public release of Menglik et al. [27] provides 4000 random-wall pairs at $\varepsilon = 1.2$, a 200-pair in-distribution test split, a 200-pair out-of-distribution split with more complex wall dials, and 5000 + 200 classical cavity pairs. The 2:1 cavity and the $Pr = 4.38$ data are generated here with the same solver.

Two splits are used throughout: extrapolation (train $\varepsilon \le 1.3$, test $\varepsilon > 1.3$, $n = 218$) and interpolation (stratified 2:1 in ε, $n = 200$). Errors are per-channel relative L_2 against Y; the floor is $\hat{Y} = X$. Training uses three seeds unless stated otherwise. When a model trained on one data set is applied to another, the target is scaled by the source per-channel standard deviation; renormalising by the target's own standard deviation changes the temperature channel by 21% and is not used.

4. Results

4.1. Field accuracy is not heat accuracy

We first ask whether the metric used to select these surrogates is the right one. Table 2 and Fig. 2 compare, for the same 200 held-out classical-cavity samples at $\varepsilon = 1.2$, the per-sample relative L_2 error of the temperature field with the per-sample error of the integral Nusselt number. The models are the ones of Sec. 4.2: a U-Net trained on the 4000 public random-wall pairs at $\varepsilon = 1.2$ and applied without retraining to the classical-cavity split.

The field-only prior reduces the median field error from 8.4% (returning X) to 0.09%. Its median integral-Nu error is 1.4–2.9% across the three seeds, against 0.48% for returning X, and the fraction of samples with a field error below 1% and a Nu error above 2% is 43% on average (23–82% across seeds). Within a seed the two errors are anti-correlated (Spearman −0.73 for one seed; the plotted points are three-seed averages): the samples whose field looks best are not the samples whose heat transfer is best. The wall-flux penalty moves the same model back into the low-error region under the six-point stencil: median Nu error 0.01%, local wall-flux error below 0.1%, and field error 0.10%.

The pattern is not an artefact of crossing boundary-condition families. The full held-out random-wall split has 200 samples, of which 22 lie at $\varepsilon = 1.1$–1.3. On those 22, a prior trained on the other 400 random-wall samples has a 0.47% median temperature-field error and a 4.04% median integral-Nu error. The wall-flux penalty reduces the median Nu error to 0.58% and the median local flux error from 4.94% to 1.14%, while the median

field error moves from 0.47% to 0.49%. On average across five seeds, 66% of these samples (50–77% across seeds) have the field error below 1% and the Nu error above 2%. The field norm is insensitive to the wall gradient in both families.

Table 2: Field error versus wall heat transfer on 200 held-out classical-cavity samples at $\varepsilon = 1.2$. T is the per-sample median relative L_2 error; the Nu columns are the hot wall, cold wall, and both walls normalised by the true total; the last two columns are the mean absolute wall-flux profile error and the fraction of samples with $T < 1\%$ and $Nu > 2\%$. Values are means across three seeds; for the field-only prior the both-wall error ranges from 1.4 to 2.9% and the fraction from 23 to 82%. The wall-flux row uses the penalty of Eq. (18).

model	T median	Nu hot	Nu cold	Nu both	local flux	fraction
return X	8.4%	0.67%	0.64%	0.48%	5.6%	0%
field-only prior	0.09%	2.0%	2.1%	2.0%	4.5%	43%
prior + wall-flux	0.10%	0.01%	0.01%	0.01%	0.08%	0%

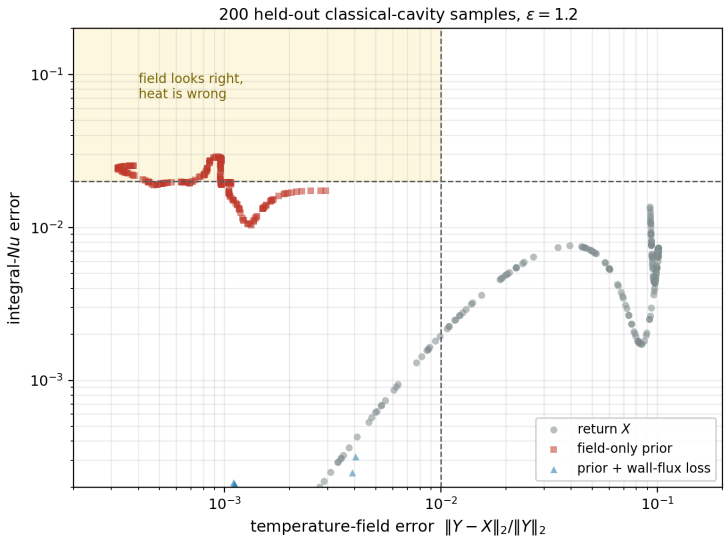


Figure 2: Per-sample temperature-field error against integral-Nu error for the 200 held-out classical-cavity samples at $\varepsilon = 1.2$. The shaded region is where a model looks accurate in the field but is wrong in heat transfer. The field-only prior sits there; the wall-flux penalty moves it out. Points are averages over three seeds. The prior's median Nu error is 1.4–2.9% across seeds; the 43% entry is the mean of the per-seed fraction, which spans 23–82%.

The mechanism is visible in the correction itself. The compressible member adds heat over the lower half of the cavity and removes it over the upper half; the integral Nu is the Nu-weighted mean of this redistribution and al-

most cancels, while the wall-normal gradient changes by several per cent. A field norm dominated by the interior does not see this boundary-layer bias. Section 4.3 gives the per-wall breakdown, the stencil dependence and the Rayleigh dependence.

4.2. A single-contrast data set becomes reusable

Table 3 and Fig. 3 show the central result. A plain U-Net trained on the public 4000 random-wall pairs at $\varepsilon = 1.2$ reaches 0.142 in u_x on our extrapolation split, barely better than returning X (0.3018). Applying the known scaling to its output, $\hat{Y} = X + (\varepsilon/\varepsilon_0)N_{\text{plain}}(X)$, reaches 0.091. Training the same architecture on D/ε reaches 0.091, within a few per cent. The 1.55-fold gain over the unscaled plain model therefore comes from applying ε, not from the network parameterization. The prior's value is that it makes the scaling explicit and can be trained without ε as an input.

Channel-averaged, the error falls from 0.129 to 0.085, a factor 1.5. Relative to returning X, the channel average is 3.1× lower (3.3× in u_x). The public prior is better than the same prior trained on our own 382 samples (0.1250), because it sees ten times more configurations.

Table 3: Transfer from a single-contrast public data set to our $\varepsilon \in [0.2, 1.9]$ samples. Entries are relative L_2 errors for $u_x/u_y/\theta/p$. Both public sets are at $\varepsilon \equiv 1.2$. The "own data" row uses 382 training samples for extrapolation and 400 for interpolation, with the P3 prior (150 epochs, one seed). The "plain, rescaled" row applies $\hat{Y} = X + (\varepsilon/\varepsilon_0)N_{\text{plain}}(X)$ to the plain model's output; it is the fair baseline for the prior. The public 4000 rows use three seeds; the 5000 rows use one, so differences below 0.003 there are not resolved. The 0.2% quoted in the text is the difference between the "plain, rescaled" and "prior" rows (0.0912 vs 0.0914), not their absolute error.

training data	model	split	u_x	u_y	θ	p
do nothing	$\hat{Y} = X$	extrap.	0.3018	0.3731	0.0556	0.3320
		interp.	0.2203	0.2683	0.0381	0.2386
public 4000, random walls	plain	extrap.	0.1417	0.1717	0.0225	0.1811
		interp.	0.1195	0.1437	0.0189	0.1422
public 4000, random walls	plain, rescaled	extrap.	0.0912	0.1081	0.0118	0.1287
		interp.	0.0649	0.0766	0.0078	0.0871
public 4000, random walls	prior	extrap.	0.0914	0.1080	0.0118	0.1287
		interp.	0.0651	0.0767	0.0077	0.0872
own data, random walls	prior	extrap.	0.1250	0.1321	0.0170	0.1719
		interp.	0.0791	0.0811	0.0103	0.0947
public 5000, classical cavity	plain	extrap.	0.3250	0.4007	0.0542	0.3167
		interp.	0.2937	0.3961	0.0555	0.3251
public 5000, classical cavity	prior	extrap.	0.4078	0.4898	0.0715	0.3222
		interp.	0.3071	0.3643	0.0532	0.2487

Applying the scaling also changes the failure mode. The plain model's error at an unseen contrast grows with the distance from the training con-

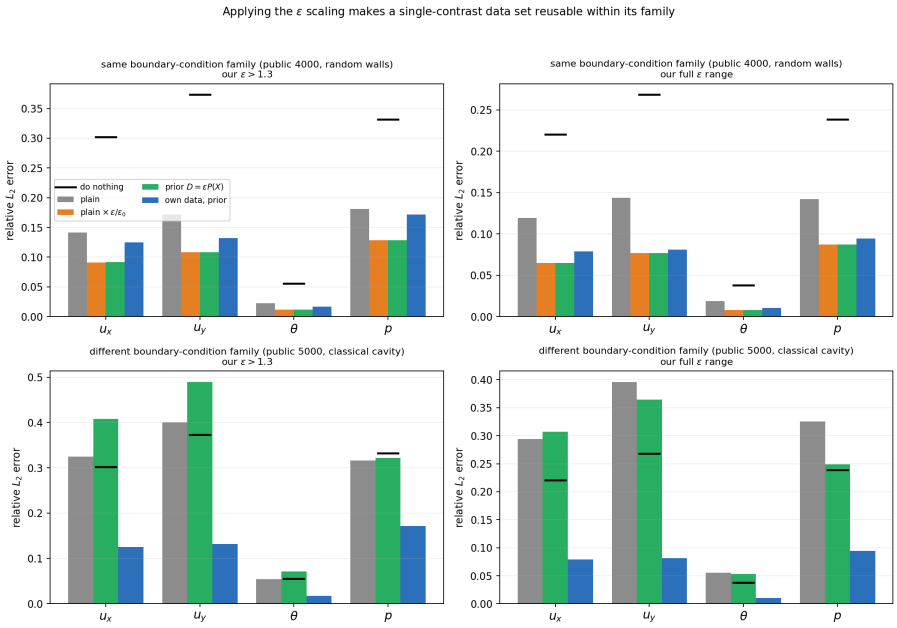


Figure 3: Reusing a single-ε data set. Within the boundary-condition family the prior transfers and beats the unscaled plain model; a plain model with the same scaling applied outside agrees with the factorized target to 0.2% relative (0.0912 vs 0.0914; Table 3). Across families the prior fails and can be worse than returning X. The failure is the flow dependence of A, not the ε scaling. Public-data models are applied with the source per-channel normalisation.

trast, because its correction is a fixed field. The rescaled model's error stays proportional to the error in P, roughly uniformly in ε.

The paired scan makes the information limit explicit. Within one configuration X is identical for all five contrasts, so an ε-blind model must emit one field for all five. At $\varepsilon = 1.9$ the error is 0.388/0.575/0.060/0.424 for $\hat{Y} = X$, 0.243/0.295/0.031/0.273 for a model trained only at $\varepsilon = 1.2$, and 0.209/0.268/0.028/0.235 for the unweighted ε-mean field ($u_x/u_y/\theta/p$, median over 20 configurations). No function of X alone can beat the best single field. The missing input is the scalar ε, not network capacity.

Across boundary-condition families the picture is different. Trained on the 5000 classical cavity pairs, the plain model lands at 0.3250 in u_x, worse than the floor of 0.3018, so it learns little that transfers. The prior is worse in extrapolation (0.4078/0.4898/0.0715/0.3222) and does not beat returning X in interpolation on any channel (u_x, u_y, θ or p; 0.3071/0.3643/0.0532/0.2487). This is the predicted boundary: the prior carries the ε dependence, not another family's prefactor. The classical cavity's peak Boussinesq speed is 41% above the random-wall median, so $\hat{u}$ separates the families; it does not order the per-sample error within a family (Spearman −0.20). It is a

family-level diagnostic, not a within-family transfer law.

The reverse direction is favorable for the field. The 4000 random-wall set contains the one-mode classical cavity as a special case. The broad-family prior reaches 0.09% relative L_2 on the 200 classical-cavity samples at $\varepsilon = 1.2$, and its wall-flux variant 0.10% (Table 2); the field-only prior's both-wall Nu error still spans 1.4–2.9% across seeds (2.0% seed mean). The asymmetry is coverage, not a violation of the law: a model trained on a narrow family cannot cover a broader one, while a model trained on a broad family can cover a subfamily. This is the practical rule for choosing the training set.

The accuracy trade-off is visible when ε is available at training time. On our own data, the ε-input network is more accurate in extrapolation than the prior; removing ε from the input costs accuracy but buys the ability to apply the model at another contrast. The gap is the price of reuse. At small sample sizes the structured model is also the cheaper one: at $n = 50$ the prior is 1.4× better than the ε-input network in u_x (0.1308 against 0.1831), and it remains ahead at $n = 200$; the crossover is near $n \approx 300$ for u_x and u_y, and θ never crosses. The prior at $n = 200$ reaches what the ε-input network needs about 270–290 samples to reach.

4.3. Wall heat transfer: per-wall breakdown and Rayleigh dependence

Table 2 splits the integral error by wall and adds the local profile error. The hot-wall integral change between the two members is 0.67% (cold wall 0.64%, both walls 0.48%; mean per-sample hot-wall ratio 0.54%, median 0.58%), while the local wall flux is redistributed by 4.25% in L_1 (p90 7.33%, pointwise excursions ±10%). The 4.25% is the L_1 norm of the relative local change $\Delta Nu(y)/Nu(y)$ between X and Y. The 5.6% entry in Table 2 (6.0/5.2% on the two walls) is the mean absolute error of the predicted wall-flux profile divided by its mean absolute value; the two use different denominators and are not directly comparable. The both-wall 2.0% error is the flux-weighted combination of the per-wall 2.0% (hot) and 2.1% (cold) values.

The public prior reduces the median global temperature-field error from 8.4% to 0.09%, but its both-wall integral Nu error is 2.0% (1.4–2.9% across seeds), larger than the 0.48% of returning X. The wall-normal derivative carries a bias that the field norm does not see. Adding the wall-flux penalty brings the six-point integral error to 0.01% and the local flux error to 0.08%, at a cost of 0.01 percentage point in the median field error (0.09% to 0.10%). With four- and eight-point stencils the same models give 0.3–0.9%, against 1.8–3.6% for the field-only prior and 0.4–0.9% for returning X; the ordering is stable, the absolute value depends on the stencil. At $\varepsilon = 0.4/0.8/1.6$ the penalty gives 0.17/0.20/1.21% against 0.45/1.08/4.08% for returning X; at $\varepsilon = 1.9$ the gain is not resolved.

The Rayleigh dependence is equally important. On the five isothermal-sidewall configurations, the integral change at $\varepsilon = 1.2$ spans +0.01 to +0.87%.

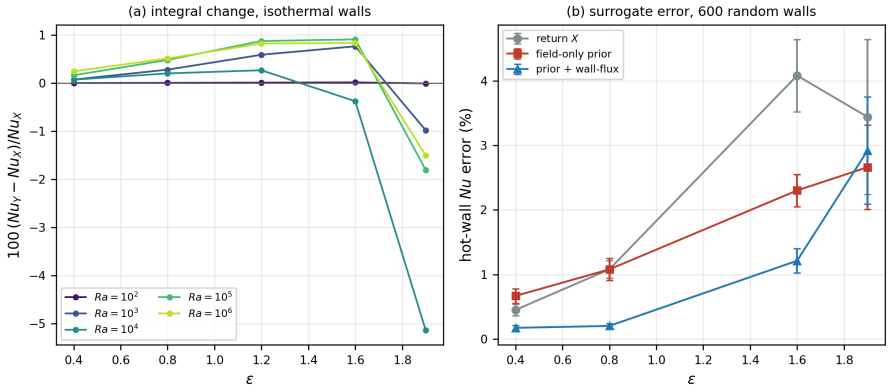


Figure 4: (a) Integral-Nu change between the two members for the five isothermal-wall configurations at $Ra = 10^2$–10^6. The change is not a single function of ε: it peaks at −5.1% at $Ra = 10^4$ and is −1.5 to −1.8% at higher Ra. (b) Hot-wall Nu error of the surrogate on the 600 random-wall samples. The wall-flux penalty reduces the error at intermediate contrast; at $\varepsilon = 1.9$ the improvement is not resolved.

At $\varepsilon = 1.9$ the same five configurations give −0.01, −0.98, −5.14, −1.81 and −1.50% for $Ra = 10^2$, 10^3, 10^4, 10^5 and 10^6 (64^2 values): the change peaks at $Ra = 10^4$ and is −1.5 to −1.8% at higher Ra. A model calibrated on Nu alone does not transfer across Ra; the local field carries the correction. At $Ra = 10^6$ the 256^2 Boussinesq solution gives $Nu = 8.793$ on the hot wall, 0.36% below the benchmark 8.825 of Le Quéré [28]; the 64^2 and 128^2 values are 8.563 and 8.759.

Grid refinement changes the size of the compressible correction (Table 4). For the $c16$ configuration at $Ra = 10^6$, the hot-wall ΔNu at $\varepsilon = 1.2$ is +0.82% at 64^2, +0.56% at 128^2 and +0.44% at 256^2, with a Richardson limit of +0.40%. At $\varepsilon = 1.9$ the same sequence gives −1.50%, −1.88% and −2.10% at 64^2, 128^2 and 192^2. The 64^2 target therefore overstates the $\varepsilon = 1.2$ change by about 0.4 percentage points and understates the $\varepsilon = 1.9$ change by about 0.6 points. The scaling exponent is unaffected: the $64^2/128^2$ ratio of R stays within 1% up to $\varepsilon = 1.2$ and within 3.4% at $\varepsilon = 1.9$ across $Ra = 10^4$–10^6.

Table 4: Grid refinement of the hot-wall $\Delta Nu = (Nu_{\mathrm{comp}} - Nu_{\mathrm{B}})/Nu_{\mathrm{B}}$ for the $c16$ isothermal-sidewall configuration at $Ra = 10^6$. The 256^2 pair at $\varepsilon = 1.9$ did not converge, and the 192^2 run was made only at $\varepsilon = 1.9$. The 64^2 column is the target against which all surrogate errors in this paper are measured.

ε	64^2	128^2	192^2	256^2
1.2	+0.82%	+0.56%	—	+0.44%
1.9	−1.50%	−1.88%	−2.10%	—

4.4. Cost

Table 5 gives the wall-clock cost. A paired solve splits into a Boussinesq member (median 11.7 s) and a compressible member (22.3 s). The surrogate replaces only the second. Per pair the cost falls from 33.9 s to 11.7 s, a factor 2.9. For a single flow swept over 100 contrasts, one Boussinesq solve plus 100 inferences costs 16.6 s against 3.77×10^3 for 100 compressible solves, a factor 2.3×10^2. For the 20-flow × 5-contrast scan, the same-reuse comparison gives 4.07×10^3 s against 321 s, a factor 13. That scan's Boussinesq median is 16.0 s, so 20×16.0 s $+100 \times 14$ ms reproduces the 321 s total. Inference takes 6.3 ms on a GPU and 14.0 ms on one CPU thread.

Table 5: Wall-clock cost on the 64^2 mesh. Solver rows use eight CPU threads; the surrogate runs on an RTX 4060 Laptop GPU (batch 1) or one CPU thread. The 100-sample scan rows use that scan's own timings; medians are not additive. In the 20-flow scan the paired median is 40.7 s per point, so the 100-point baseline is 4.07×10^3 s; its Boussinesq median is 16.0 s, consistent with 20×16.0 s $+100 \times 14$ ms = 321 s. The 20-flow row with reuse includes 20 Boussinesq solves and 100 inferences; its baseline repeats both members at all 100 points.

quantity	median	mean	p90	max
paired solve, 600 samples	33.9 s	38.0 s	59.9 s	510 s
Boussinesq member alone	11.7 s	13.1 s	21.7 s	—
compressible member alone	22.3 s	25.0 s	37.3 s	—
Boussinesq member, 100-sample scan	16.0 s	—	—	—
compressible member, 100-sample scan	37.5 s	—	—	—
paired solve, 100-sample ε scan	55.8 s	67.7 s	136 s	202 s
surrogate, GPU, batch 1	6.3 ms	—	—	—
surrogate, GPU, batch 256	0.49 ms/sample	—	—	—
surrogate, CPU, 1 thread	14.0 ms	—	—	—
end-to-end pair (Boussinesq + surrogate)	11.7 s	—	—	—
20-flow × 5-contrast scan, paired baseline	4.07×10^3 s	—	—	—
same scan, one Boussinesq per flow + 100 inferences	321 s	—	—	—

4.5. Why the scaling is not a fitting artefact

Three independent checks separate the scaling from a flexible network that happened to fit the data. Mesh refinement and solver-noise checks are reported in the Supplementary Material; they do not change the exponent or the sign of the correction.

First, a Fourier neural operator with 1.19M parameters shows the same pattern. Against its ε-blind version, u_x falls from 0.2287 to 0.1334 with the scaling and ε multiplied outside, a factor 1.71. It is statistically indistinguishable from an explicit ε input (0.1334 against 0.1342). On u_y and θ the

prior is better; only on p does the explicit input win (0.1536 against 0.1734). The scaling is therefore not tied to a particular backbone.

Second, a POD-ROM with at most four modes on the 15 three-dimensional samples at $Ra = 10^4$ (part of the 33 pairs) improves from 0.168 to 0.081 in u_x when it predicts D/ε instead of D; u_y, θ and p improve by 1.27, 1.25 and 1.67. The gain is largest at small contrast and is not uniform.

Third, one Newton step of the low-Mach residual at the Boussinesq state, on the same discretisation, reproduces the measured correction to median 4.7/3.8/7.2% in $u_x/u_y/p$ on 27 samples at $\varepsilon = 0.02$–0.10, with no extrapolation in ε. The extrapolated P predicts held-out measurements at $\varepsilon = 0.1$ and 0.2 to 2.4/1.8/3.8%. This is the linearised operator doing the work, not an extra fit.

5. Discussion

5.1. What the result does and does not claim

The result is a model-selection rule plus a scaling law, not a universal surrogate. The selection rule is the one that matters for heat transfer: relative L_2 field error can rank a model above doing nothing on integral Nu. The scaling law explains why the two can disagree: the correction is largely a wall-flux redistribution that almost cancels in the integral. The law itself is stable across geometry and fluid, and a 33-pair three-dimensional check is consistent with it; the prefactor is a functional of the flow. Across the three 3-D wall types it varies by factors of 2–13, and changing the fluid from $Pr = 0.71$ to 4.38 moves it by 3.3% for the velocity and temperature channels and 6.1% for pressure. A surrogate that takes X as input can learn the prefactor; a lookup table of A cannot.

This also explains the cross-family failure. The law transfers the ε scaling, which is universal within the retained equations, and not the flow-dependent amplitude. For an engineering application the practical condition is therefore: train on the boundary-condition family and Ra range of interest, then reuse the same model across contrasts without new compressible solves. That is a weaker claim than "train anywhere and apply everywhere", and it is the claim the data support.

5.2. Relation to existing methods

To our knowledge, the closest prior work either learns a correction field at a fixed contrast or treats the contrast as an input channel. Neither yields a scaling law for the gap or a way to reuse a single-contrast data set. Residual-based corrector operators solve a linear variational problem for the residual of a trained operator and add the solution to its prediction [29, 30]. They are accurate but cost one linear solve per query. The present rule is a single

multiplication outside the network; the linear solve is used once, offline, to identify P. Physics-informed extrapolation adds sparse observations or residual terms [31], and transfer operator learning addresses conditional shift [32]. Multi-fidelity surrogates combine cheap and expensive data through learned corrections [33, 34, 35]; they still learn the parameter dependence from samples. The difference here is that the parameter dependence is derived and only the spatial shape is learned.

Dimension-informed operator learning imposes exact units equivariance and proves invariance under rescaling [36]. That is a different invariance: it concerns changes of units, not the physical breaking of the Boussinesq similarity by a finite temperature contrast. The present law is specific to the Boussinesq–compressible pair and predicts the form of the correction, not only its dimensional consistency.

5.3. Practical recipe

For a new flow family:

1. solve the Boussinesq member X on the target family;
2. obtain paired data at one accessible contrast ε_0;
3. train N on $P = D/\varepsilon_0$ with the residual form $\hat{Y} = X + \varepsilon N(X)$, or train a plain model at ε_0 and rescale its output by $\varepsilon/\varepsilon_0$;
4. if wall heat transfer matters, add the wall-flux penalty of Eq. (18);
5. apply at other contrasts without new compressible solves, keeping the family and Ra range fixed.

The cost of the single paired data set is paid once. The model then replaces the compressible member at every other contrast. If the target family is outside the training family, the prefactor must be recalibrated; the ε scaling itself does not need to be relearned. The two implementations above agree to 0.2% relative on the extrapolation split, 0.0912 against 0.0914 (Table 3). The reusable element is the external ε factorisation, not the residual parameterisation or a network-free scaling law.

5.4. Scope

The main sweep is two-dimensional and steady, with $\varepsilon < 2$ and $Ma \lesssim 2 \times 10^{-3}$. The three-dimensional check uses 33 pairs at one resolution ($res = 16$) and three Rayleigh numbers; it supports the exponent, not a three-dimensional design study. The pressure channel is quoted in a fixed gauge. The temperature channel has a flow-dependent amplitude and is not part of the common-exponent claim. All surrogate field and Nu errors are relative to the 64^2 compressible target Y. That target is not grid-converged: at $Ra = 10^6$ the hot-wall ΔNu changes from 0.82% (64^2) to 0.44% (256^2) at $\varepsilon = 1.2$, and from −1.50% to −2.10% at $\varepsilon = 1.9$ (192^2; Table 4). The 0.01%

value is therefore an error against the 64^2 target, not an absolute physical accuracy. It also uses the six-point stencil of the training loss; the four- and eight-point evaluations are reported above. These limits define the next experiments.

6. Conclusion

Relative L_2 field error is not a sufficient selection metric for a heat-transfer surrogate. On 200 held-out classical-cavity samples, a model with a 0.09% median temperature-field error missed the both-wall integral Nusselt number by 2.0% (1.4–2.9% across seeds); in 43% of the samples on average (23–82% across seeds) the field error was below 1% and the Nu error above 2%. On the 22 random-wall samples at $\varepsilon = 1.1$–1.3, the corresponding values are 0.47% and 4.04%, with 66% on average across five seeds (50–77%) in that region. Under the six-point training stencil, a wall-flux penalty reduced the both-wall error to 0.01% and 0.58%, respectively, and the local wall-flux error to 0.08% and 1.14%; four- and eight-point stencils give 0.3–0.9%, overlapping returning X (0.4–0.9%). The median field error moved to 0.10% and 0.49%.

The gap between Boussinesq and compressible steady convection factorises. The relative temperature contrast sets its leading-order amplitude, and the flow sets its spatial shape. Across 627 paired solutions, the velocity channels follow $R = A\varepsilon^m$ with exponents indistinguishable from one; a quadratic refinement gives positive curvature $\kappa = 0.70$ and 0.83 (point estimates). The prefactor is a functional of the flow and changes with geometry and boundary conditions, and weakly with the fluid (3.3% for velocity and temperature, 6.1% for pressure). A 33-sample three-dimensional check is consistent with the same exponent.

The reusable element is the external ε factorisation. A plain model trained at $\varepsilon_0 = 1.2$ and rescaled by $\varepsilon/\varepsilon_0$ agrees with the factorized target to 0.2% relative on the extrapolation split (0.0912 vs 0.0914); the residual parameterisation is a convenient implementation, not the source of the transfer. The law makes a public single-contrast data set usable within its boundary-condition family: the channel-averaged extrapolation error falls from 0.129 to 0.085, 3.1× lower than returning the Boussinesq solution. The gain is not a universal accuracy win over an ε-aware model; it is the ability to apply a model trained at one contrast at another without new compressible solves.

The combination of the scaling law and the wall-flux loss gives a surrogate that is reusable across temperature contrast; its heat-transfer accuracy is meaningful at the 0.4–0.6 percentage-point resolution set by the 64^2 target, not as an absolute wall-flux benchmark.

Supplementary material

The Supplementary Material contains the five asymptotic checks, the two-parameter (Ra, ε) fits, the exact single-contrast transfer matrix, the learning curve, the public-split statistics, the θ–$\hat{u}$ fits, the three-dimensional fit table, the POD-ROM details, the per-sample field–Nu statistics, the multi-seed transfer and wall-flux tables, mesh-independence and solver-noise checks, and the script-to-result map.

CRediT authorship contribution statement

The authors contributed to conceptualization, methodology, software, validation, formal analysis, investigation, writing (original draft), and writing (review and editing).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the authors used an AI assistant to develop analysis scripts, to draft and edit the manuscript, and to check numerical consistency between the text and the stored results. After using this tool, the authors reviewed and edited the content and take full responsibility for the content of the publication.

Data availability

The public paired data set of Menglik et al. [27] is available under CC BY 4.0. Our generated data, solver configuration and scripts that reproduce the numbers in this paper are available at https://github.com/sheep-cloud12138/Algorithms-instead-of-simulations (branch dev). The raw solver output is available from the authors on request.

References

[1] M. Giacomini, P. Díez, Surrogates for physics-based and data-driven modelling of parametric systems: Review and new perspectives, Archives of Computational Methods in Engineering 33 (6) (2026) 8519–8551. doi:10.1007/s11831-026-10552-4.

[2] C. Meng, S. Griesemer, D. Cao, S. Seo, Y. Liu, When physics meets machine learning: A survey of physics-informed machine learning, Machine Learning for Computational Science and Engineering 1 (1) (2025) 20. doi:10.1007/s44379-025-00016-0.

[3] E. A. Spiegel, G. Veronis, On the Boussinesq approximation for a compressible fluid, The Astrophysical Journal 131 (1960) 442–447. doi:10.1086/146849.

[4] D. D. Gray, A. Giorgini, The validity of the Boussinesq approximation for liquids and gases, International Journal of Heat and Mass Transfer 19 (5) (1976) 545–551. doi:10.1016/0017-9310(76)90168-X.

[5] Q. Huhn, M. Tano, J. C. Ragusa, Y. Choi, Parametric dynamic mode decomposition for reduced order modeling, Journal of Computational Physics 475 (2022) 111852. doi:10.1016/j.jcp.2022.111852.

[6] F. Feng, Y. Li, Z. Chen, W.-T. Wu, J.-Z. Peng, M. Mei, Graph convolution network-based surrogate model for natural convection in annuli, Case Studies in Thermal Engineering 57 (2024) 104330. doi:10.1016/j.csite.2024.104330.

[7] J.-Z. Peng, N. Aubry, Y. Li, M. Mei, Z. Chen, W.-T. Wu, Physics-informed graph convolutional neural network for modeling geometry-adaptive steady-state natural convection, International Journal of Heat and Mass Transfer 216 (2023) 124593. doi:10.1016/j.ijheatmasstransfer.2023.124593.

[8] L. Lu, P. Jin, G. Pang, Z. Zhang, G. E. Karniadakis, Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators, Nature Machine Intelligence 3 (3) (2021) 218–229. doi:10.1038/s42256-021-00302-5.

[9] S. Wang, H. Wang, P. Perdikaris, Learning the solution operator of parametric partial differential equations with physics-informed DeepONets, Science Advances 7 (40) (2021) eabi8605. doi:10.1126/sciadv.abi8605.

[10] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, A. Anandkumar, Fourier neural operator for parametric partial differential equations, in: International Conference on Learning Representations, 2021, arXiv:2010.08895.

[11] X. Zhao, X. Chen, Z. Gong, W. Zhou, W. Yao, Y. Zhang, RecFNO: A resolution-invariant flow and heat field reconstruction method from sparse observations via Fourier neural operator, International Journal of Thermal Sciences 195 (2023) 108619. doi:10.1016/j.ijthermalsci.2023.108619.

[12] X. F. Yu, W. Gao, K. Guo, Z. Zhang, H. Wang, M. Wang, S. Qiu, W. Tian, Fourier neural operator-based approach for real-time prediction of high-temperature heat pipes frozen startup, Applied Thermal Engineering 302 (2026) 131964. doi:10.1016/j.applthermaleng.2026.131964.

[13] M. Raissi, P. Perdikaris, G. E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, Journal of Computational Physics 378 (2019) 686–707. doi:10.1016/j.jcp.2018.10.045.

[14] S. Cai, Z. Wang, S. Wang, P. Perdikaris, G. E. Karniadakis, Physics-informed neural networks for heat transfer problems, ASME Journal of Heat and Mass Transfer 143 (6) (2021) 060801. doi:10.1115/1.4050542.

[15] D. Jalili, S. Jang, M. Jadidi, G. Giustini, A. Keshmiri, Y. Mahmoudi, Physics-informed neural networks for heat transfer prediction in two-phase flows, International Journal of Heat and Mass Transfer 221 (2023) 125089.

doi:10.1016/j.ijheatmasstransfer.2023.125089.

[16] I. Şahin, C. Moya, A. Mollaali, G. Lin, G. Paniagua, Deep operator learning-based surrogate models with uncertainty quantification for optimizing internal cooling channel rib profiles, International Journal of Heat and Mass Transfer 219 (2023) 124813. doi:10.1016/j.ijheatmasstransfer.2023.124813.

[17] T. De Ryck, S. Mishra, Numerical analysis of physics-informed neural networks and related models in physics-informed machine learning, Acta Numerica 33 (2024) 633–713. doi:10.1017/S0962492923000089.

[18] K. Luo, J. Zhao, Y.-P. Wang, J. Li, J. Wen, J. Liang, H. Soekmadji, S. Liao, Physics-informed neural networks for PDE problems: A comprehensive review, Artificial Intelligence Review 58 (10) (2025) 323. doi:10.1007/s10462-025-11322-7.

[19] H. Yik, V. Valori, S. Weiss, Turbulent Rayleigh–Bénard convection under strong non-Oberbeck–Boussinesq conditions, Physical Review Fluids 5 (10) (2020) 103502. doi:10.1103/PhysRevFluids.5.103502.

[20] M. Macek, G. Zinchenko, V. Musilová, P. Urban, J. Schumacher, Assessing non-Oberbeck–Boussinesq effects of convection in cryogenic helium, Physical Review Fluids 8 (9) (2023) 094606. doi:10.1103/PhysRevFluids.8.094606.

[21] X. Pan, J.-I. Choi, Non-Oberbeck–Boussinesq effects on a water-filled differentially heated vertical cavity, Physics of Fluids 35 (11) (2023) 113609. doi:10.1063/5.0173049.

[22] S.-N. Xia, Z.-H. Wan, S. Liu, Q. Wang, D.-J. Sun, Flow reversals in Rayleigh–Bénard convection with non-Oberbeck–Boussinesq effects, Journal of Fluid Mechanics 798 (2016) 628–642. doi:10.1017/jfm.2016.338.

[23] G. Ahlers, S. Großmann, D. Lohse, Heat transfer and large scale dynamics in turbulent Rayleigh–Bénard convection, Reviews of Modern Physics 81 (2) (2009) 503–537. doi:10.1103/RevModPhys.81.503.

[24] J. Zhang, Y. Wu, N. Gui, X. Yang, J. Tu, S. Jiang, Investigation of compressible flow under low Mach number in an enclosed square cavity with a novel non-Boussinesq algorithm, Physics of Fluids 35 (12) (2023) 123120. doi:10.1063/5.0175738.

[25] H. Satake, T. Tagawa, Mathematical modeling for low-Mach-number compressible thermal convection analysis in a square enclosure, Physics of Fluids 36 (12) (2024) 127168. doi:10.1063/5.0243786.

[26] P. Bella, E. Feireisl, F. Oschmann, Rigorous derivation of the Oberbeck–Boussinesq approximation revealing unexpected term, Communications in Mathematical Physics 403 (3) (2023) 1245–1273. doi:10.1007/s00220-023-04823-5.

[27] N. Menglik, A. Shao, D. Hyde, A neural surrogate approach for simulating natural convection problems (2026). arXiv:2606.25259.

[28] P. Le Quéré, Accurate solutions to the square thermally driven cavity at high Rayleigh number, Computers & Fluids 20 (1) (1991) 29–41. doi:10.1016/0045-7930(91)90025-D.

[29] P. K. Jha, Residual-based error corrector operator to enhance accuracy and reliability of neural operator surrogates of nonlinear variational boundary-value problems, Computer Methods in Applied Mechanics and Engineering 419 (2023) 116595. doi:10.1016/j.cma.2023.116595.

[30] L. Cao, T. O'Leary-Roseberry, P. K. Jha, J. T. Oden, O. N. Ghattas, Residual-

based error correction for neural operator accelerated infinite-dimensional Bayesian inverse problems, Journal of Computational Physics 486 (2023) 112104. doi:10.1016/j.jcp.2023.112104.

[31] M. Zhu, H. Zhang, A. Jiao, G. E. Karniadakis, L. Lu, Reliable extrapolation of deep neural operators informed by physics or sparse observations, Computer Methods in Applied Mechanics and Engineering 412 (2023) 116064. doi:10.1016/j.cma.2023.116064.

[32] S. Goswami, K. Kontolati, M. D. Shields, G. E. Karniadakis, Deep transfer operator learning for partial differential equations under conditional shift, Nature Machine Intelligence 4 (12) (2022) 1155–1164. doi:10.1038/s42256-022-00569-2.

[33] P. Conti, M. Guo, A. Manzoni, J. S. Hesthaven, Multi-fidelity surrogate modeling using long short-term memory networks, Computer Methods in Applied Mechanics and Engineering 404 (2022) 115811. doi:10.1016/j.cma.2022.115811.

[34] W. Chen, P. Stinis, Feature-adjacent multi-fidelity physics-informed machine learning for partial differential equations, Journal of Computational Physics 498 (2023) 112683. doi:10.1016/j.jcp.2023.112683.

[35] Y. Lyu, X. Zhao, Z. Gong, X. J. Kang, W. Yao, Multi-fidelity prediction of fluid flow based on transfer learning using Fourier neural operator, Physics of Fluids 35 (7) (2023) 077118. doi:10.1063/5.0155555.

[36] Y. Song, Y. Wu, Y. Wang, X. Yang, DimINO: Dimension-informed neural operator learning (2024). arXiv:2410.05894.